

Median change in property

VAE	-0.23	-0.23	-0.01	-0.01	-0.24	-0.24
DDPM	-0.33	-0.33	0.29	0.29	-0.11	-0.11
LDM	-0.07	-0.07	-0.95	-0.95	-0.24	-0.24
FM	-0.34	-0.34	0.32	0.32	-0.10	-0.10
DiffAb	0.06	0.11	-0.21	-0.21	-0.05	-0.04
IgLM	0.24	0.01	-0.57	-0.83	0.63	-0.51
NOS-C	0.09	-2.25	0.03	-1.07	0.08	0.10
NOS-D	0.18	-0.29	0.31	0.38	-0.09	-0.10
OAE + gradient ascent	-0.01	-0.19	-1.21	-0.33	0.43	-0.05
OAE + MCMC	-0.11	-0.19	-0.20	-0.37	-0.19	-0.10
OAE + hill climbing	0.40	-0.29	0.76	-0.87	0.53	-0.06
OAE + stochastic hill climbing	-0.12	-0.15	0.32	-0.80	0.19	-0.05
RNAGenScape (ours)	0.54	-2.81	0.77	-1.29	0.63	-0.58

Improvement +35.0 % +24.9 % +1.3 % +20.6 % +0.0 % +13.7 %

OpenVaccine (+)
OpenVaccine (-)
Zebrafish (+)
Zebrafish (-)
Ribosome (+)
Ribosome (-)

Success rate

42.1 %	57.9 %	49.4 %	50.6 %	41.7 %	58.3 %
33.8 %	66.2 %	58.2 %	41.8 %	45.9 %	54.1 %
47.5 %	52.5 %	22.5 %	77.5 %	41.8 %	58.2 %
32.5 %	67.5 %	59.8 %	40.2 %	46.4 %	53.6 %
55.0 %	44.0 %	36.3 %	60.8 %	45.0 %	54.2 %
63.1 %	49.6 %	32.0 %	74.3 %	80.5 %	65.5 %
54.6 %	90.6 %	51.1 %	80.8 %	53.6 %	46.0 %
58.3 %	65.4 %	57.9 %	40.2 %	46.5 %	53.6 %
51.0 %	61.3 %	18.0 %	66.5 %	73.6 %	55.7 %
45.1 %	57.7 %	44.3 %	60.1 %	43.0 %	54.2 %
69.2 %	64.2 %	74.4 %	75.2 %	83.0 %	56.1 %
43.5 %	60.0 %	60.3 %	73.7 %	60.4 %	55.2 %
77.5 %	97.9 %	75.0 %	85.1 %	81.4 %	67.8 %

+12.0 % +8.1 % +0.8 % +5.3 % -1.9 % +3.5 %

OpenVaccine (+)
OpenVaccine (-)
Zebrafish (+)
Zebrafish (-)
Ribosome (+)
Ribosome (-)